Lab 2

Week 2

AMBA AHB

Implement a simple AHB-Lite master/slave and test read/write transactions.

AMBA AHB

Course: Industrial Protocols
Prerequisite: Students should already understand ready/valid handshaking, master/slave roles, and the APB two-phase (SETUP/ACCESS) transfer model.

Recap: Where APB Hits a Wall

"APB takes a minimum of 2 cycles per transfer, with zero overlap between transfers. If your CPU needs to fetch an instruction every cycle, what happens?"

It can't. Every single APB transfer is fully atomic — address phase and data phase never overlap across transfers. For a peripheral like a UART that transfers a byte every few thousand cycles, this is irrelevant. For something like on-chip SRAM that a CPU fetches from every cycle, it's a hard bottleneck.

The fix is not "make APB faster." APB's simplicity is the *point* — you don't want to add pipelining complexity to a UART register interface. Instead, AMBA offers a **different protocol** for exactly this higher-bandwidth need: AHB.

What Pipelining Means Here

In APB, the transfer N's address phase and data phase happen, then (at earliest) transfer N+1 begins. In AHB:

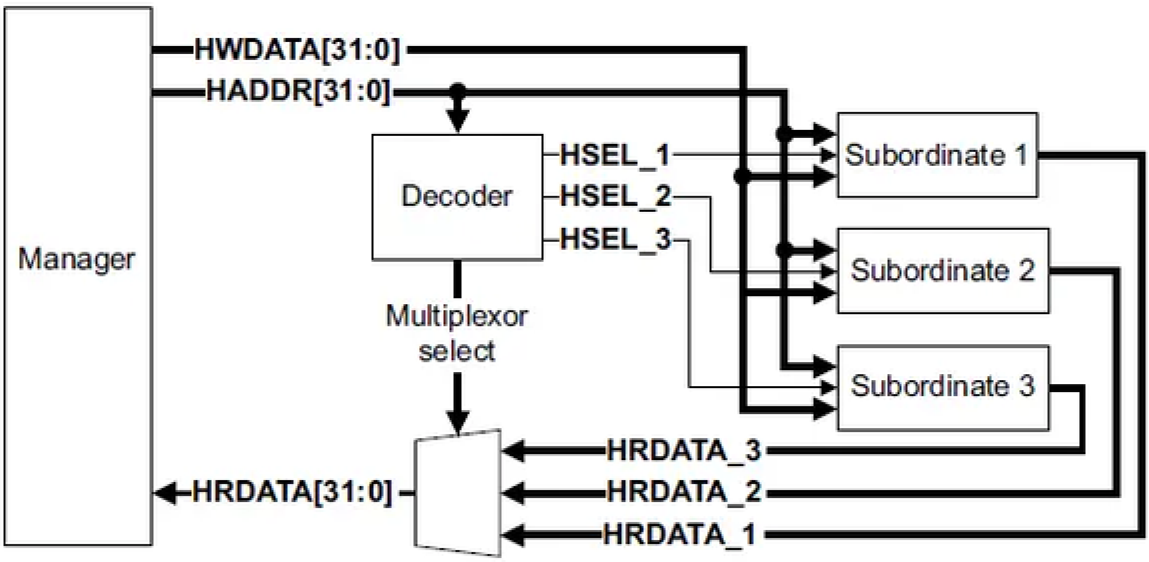
APB (non-pipelined):
Transfer 1: [Addr] [Data]
Transfer 2:   [Addr] [Data]
Transfer 3:   [Addr] [Data]
-> 2 cycles minimum per transfer, zero overlap

AHB (pipelined):
Transfer 1: [Addr]
Transfer 2:   [Addr]
Transfer 3:   [Addr]
Transfer 1:   [Data]
Transfer 2:   [Data]
Transfer 3:   [Data]
-> address phase of transfer N+1 overlaps data phase of transfer N
-> steady-state throughput: 1 transfer per cycle

AHB system roles

Role	Typical implementation	Main job
Manager/Master	Formerly Master; CPU bus interface, DMA engine, bridge, debug port	Active component that initiates read or write operations and drives address, control, and write data
Subordinate/Slave	Formerly Slave; SRAM, peripheral, bridge, memory controller	Passive component that responds to selected requests from a Manager and returns read data, ready, and response
Interconnect	Decoder, mux, arbiter, routing fabric	Routing and arbitration logic that manages traffic between Managers and Subordinates

AHB Block diagram



In a single-Manager AHB system, the interconnect can be simple. A decoder watches HADDR and asserts one HSELx select line. A response multiplexor returns HRDATA, HREADYOUT, and HRESP from the selected Subordinate back to the Manager.

AHB Signal Set (AHB-Lite, single master)

Signal	Direction	Width	Meaning
HCLK	—	1	Clock
HRESETn	—	1	Active-low reset
HADDR	Master → Slave	up to 32	Address for the current address-phase transfer
HWRITE	Master → Slave	1	1 = write, 0 = read
HTRANS	Master → Slave	2	Transfer type: IDLE, BUSY, NONSEQ, SEQ
HSIZE	Master → Slave	3	Transfer size: byte, halfword, word, etc.
HBURST	Master → Slave	3	Burst type/length

HWDATA	Master → Slave	up to 32	Write data (data-phase, one cycle behind HADDR)
HREADY	Slave → Master (also fed back as bus-wide signal)	1	1 = previous transfer finished this cycle; 0 = slave still busy, insert wait
HRDATA	Slave → Master	up to 32	Read data
HRESP	Slave → Master	1 (AHB-Lite)	OKAY or ERROR

Spot the new signals (HTRANS, HBURST, HSIZE); why APB never needed them (APB has no concept of transfer type/burst — every transfer is a single, uniform, complete access).

HTRANS — The Signal That Encodes the Pipeline

HTRANS is the single most important new concept this session. Four values:

HTRANS	Name	Meaning
00	IDLE	No transfer requested this cycle — bus is idle, slave should not respond
01	BUSY	Master intends more transfers in this burst, but isn't ready to supply the next address this cycle
10	NONSEQ	A single transfer, or the *first* transfer of a burst — address is unrelated to any previous one
11	SEQ	A subsequent transfer within a burst — address is the previous address + size (sequential)

HTRANS is what tells the slave "here comes a new address phase." A slave watches HTRANS every cycle to know whether to latch a new address.

HBURST — Grouping Multiple Transfers

HBURST	Meaning
000	SINGLE — one transfer only
001	INCR — incrementing burst, undefined length
010 / 011	WRAP4 / INCR4 — 4-beat burst
100 / 101	WRAP8 / INCR8 — 8-beat burst
110 / 111	WRAP16 / INCR16 — 16-beat burst

HSIZE Encoding

HSIZE tells the slave how many bytes are transferred in each beat.
The encoding is:

HSIZE[2:0]	Transfer Size	Bytes
000	Byte	1
001	Halfword	2
010	Word	4
011	Doubleword	8
100	128-bit	16
101	256-bit	32
110	512-bit	64
111	1024-bit	128

Most educational AHB-Lite designs only use the first three values:

HSIZE	Meaning
000	8-bit transfer

001	16-bit transfer
010	32-bit transfer

How HSIZE Affects Address Increment

The address increment equals the transfer size.

HSIZE	Transfer	Address Increment
000	1 byte	+1
001	2 bytes	+2
010	4 bytes	+4
011	8 bytes	+8

Lab Task

Task 1:

Design and simulate a simplified **AHB-Lite (Single Master)** bus interface in Logisim to perform read/write transfers with wait-state insertion.

Task 2:

Analyze the provided buggy AHB-Lite master-slave SystemVerilog code, develop 5 comprehensive test cases, simulate them individually, identify and document any protocol violations or functional bugs, and fix all issues to achieve correct AHB protocol operation. Provide a detailed analysis paragraph explaining the debugging process and fixes applied.

[AHB Code]

Test Case 1: Single Write Transfer (No Wait-state)
Test Case 2: Single Read Transfer (No Wait-state)
Test Case 3: Write with Wait-state Insertion
Test Case 4: Burst Transfer (INCR4 - 4 transfers)
Test Case 5: Invalid Address with Error Response

Task 3:

Develop a comprehensive block diagram of an AHB (Advanced High-performance Bus) system with 4 slaves, explain its architecture, signal interconnections, decoding logic, and operational flow in detail.

Task 4:

Identify and describe 5 high-speed peripherals and 5 low-speed peripherals commonly used in System-on-Chip (SoC) designs, with a brief description of each, including their typical data rates, bus interface requirements, and applications.